COMPETENCY STANDARD

FOR

MOBILE PHONE SERVICING

(ICT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-MPS-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-ICT-MPS-02-G	Carryout Workplace Interaction in English	1. Interpret workplace communication and etiquette 2. Read and understand workplace documents 3. Participate in workplace meetings and discussions 4. Practice professional ethics at workplace	10
		Total Hour	**20**

Sector Specific Competencies (40 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-MPS-01-S	Operate Office Application Software	1. Operate computer 2. Install application software 3. Use word processor to prepare/create documents 4. Use spreadsheet to create/ prepare worksheets 5. Use presentation software to create/prepare presentation	30
SICIP-ICT-MPS-02-S	Comply with Ethical Standards in IT Workplace	1. Uphold the requirements of clients 2. Deliver quality products and services 3. Maintain professionalism at workplace 4. Maintain workplace code of conduct	10
		Total Hour	**40**

Occupation Specific Competencies (300 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-ICT-MPS-01-O	Recognize the Basic Electrical and Electronics Concept	1. Understand basic electricity concept 2. Identify basic electronics components 3. Understand transformer functions 4. Identify low powered cables	20
SICIP-ICT-MPS-02-O	Apply Fundamental Skills for Mobile Phone Works	1. Identify basic concept of mobile phone 2. Identify types of tools and equipment used in mobile phone servicing 3. Identify the materials required in mobile phone servicing 4. Identify circuit, schematic and block diagram 5. Measure current, voltage and resistance	20
SICIP-ICT-MPS-03-O	Identify Components of Mobile Phone	1. Identify the sections of a mobile phone 2. Identify individual components and parts	10
SICIP-ICT-MPS-04-O	Disassemble and Reassemble of Mobile Phone	1. Disassemble mobile phone 2. Re-assemble the mobile phone 3. Clean tools and equipment	10
SICIP-ICT-MPS-05-O	Practice Soldering and De-soldering	1. Select soldering tools and materials 2. Perform soldering on PCB 3. Perform de-soldering components from PCB 4. Perform re-soldering from PCB 5. Clean soldering tools and workspace	60
SICIP-ICT-MPS-06-O	Test Components of Mobile Phone	1. Understand standard testing criteria for each component 2. Select correct tools and testing method 3. Test components in-circuit or off-circuit	60
SICIP-ICT-MPS-07-O	Service Mobile Phone	1. Prepare for servicing works 2. Disassemble mobile phone 3. Test mobile phone components 4. Replace faulty components 5. Re-assemble mobile phone 6. Clean tools and equipment	80

SICIP-ICT-MPS-08-O	Apply Software Installation and Flashing Mobile Phone	1. Identify software requirements of mobile phone 2. Prepare flashing tools and install drivers 3. Connect mobile phone to PC or flashing tool 4. Flash mobile phone software 5. Clean tools and equipment	40
		Total Hour	**300 Hrs.**

COMPETENCY STANDARD: MOBILE PHONE SERVICING

The Generic Competencies

Unit of Competency:	Nominal Duration:	Unit Code:
APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	10 hrs.	SICIP-ICT-MPS-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **OHS policies** and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety procedures	2.1 OHS policies and procedures are practiced and applied. 2.2 **Personal Protective Equipment (PPE)** is selected and used. 2.3 Personal hygiene is practiced.
3. Report hazards and risks	3.1 **Hazards and risks** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 **Emergency response plans and procedures** are implemented. 4.3 First aid procedure is applied during emergency situations.

Range of Variables

Variable	Range May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Bangladesh Standards for OHS Building Code 1.3 Fire Safety Rules and Regulations

2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Face mask 2.6 Goggles 2.7 Ear plugs 2.8 Scarf
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.4 Machine hazards 3.5 Materials hazards 3.6 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures – firefighting, earthquake response, explosion response & accident response 1.4 Types of hazards – biological, chemical & physical, and their effects 1.5 PPE types and uses 1.6 Personal hygiene practices 1.7 OHS awareness 1.8 Malfunctions and resolutions
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices in the workplace 2.3 Identifying, assessing and controlling hazards and risks 2.4 Reporting incidents arising from hazards and risks 2.5 Responding to emergency procedures 2.6 Maintaining physical well-being in the workplace 2.7 Implementing emergency response plans and procedures 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Learning manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied OHS policies and procedures 1.2 selected and used personal protective equipment (PPE) 1.3 practiced personal health and safety procedures 1.4 identified, assessed and controlled hazards and risks 1.5 reported incidents arising from hazards and risks to the authority 1.6 implemented plans and procedures to respond emergencies 1.7 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT WORKPLACE INTERACTION IN ENGLISH**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-ICT-MPS-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out workplace interaction in English. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions, and practicing professional ethics at workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace code of conducts are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **courteous manner** to gather and convey information. 1.4 Questions about routine **workplace procedures and matters** are asked and responded as required.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted as per standard. 2.2 Assistance is taken to aid comprehension when required from peers/supervisors. 2.3 Visual information/symbols/signages are understood and followed. 2.4 Specific and relevant information are accessed from **appropriate sources.** 2.5 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time and followed meeting procedures and etiquette. 3.2 Own opinions are expressed and listened to those of others without interruption. 3.3 Inputs are provided consistent with the meeting purpose and interpreted and implemented meeting outcomes.
4. Practice professional ethics at workplace	4.1 Responsibilities as a team member are demonstrated and kept promises and commitments made to others. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is respected and maintained. 4.4 Situations and actions considered inappropriate or which present a conflict of interest are avoided.

Range of Variables

Variable	Range May Include but not limited to:
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills
2. Workplace procedures and matters	2.1 Notes 2.2 Agenda 2.3 Simple reports such as progress and incident reports 2.4 Job sheets 2.5 Operational manuals 2.6 Brochures and promotional material 2.7 Visual and graphic materials 2.8 Standards 2.9 OSH information 2.10 Signs
3. Appropriate sources	3.1 HR Department 3.2 Managers 3.3 Supervisors

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Workplace code of conducts 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette
2. Underpinning Skills	2.1 Demonstrating performance of workplace communication and etiquette 2.2 Applying workplace information and symbols/signages 2.3 Applying workplace code of conducts as per organizational guidelines 2.4 Providing inputs in the meeting consistent with the meeting purpose 2.5 Interpreting and implementing meeting outcomes 2.6 Practicing professional ethics at workplace 2.7 Avoiding tactfully situations and actions considered inappropriate or which presents a conflict of interest
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Work book 4.3 Learning & operational manuals 4.4 Workplace tools/equipment 4.5 Relevant tools, equipment, software and facilities needed to perform the activities

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 practiced workplace code of conducts as per organizational guidelines 1.3 Interpreted workplace documents, instructions and symbols 1.4 participated in workplace meetings and discussions 1.5 interpreted and implemented meeting outcomes 1.6 practiced professional ethics at workplace
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

The Sector Specific Competencies

Unit of Competency: **OPERATE OFFICE APPLICATION SOFTWARE**	**Nominal Duration:** 30 hrs.	**Unit Code:** SICIP-ICT-MPS-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate office application software. It specifically includes the tasks of operating computer, installing application software, using word processor to prepare/create documents, using spreadsheet to create/ prepare worksheets, using presentation software to create/prepare presentation and printing a document.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Operate computer	1.1 Safe workplace practices are observed according to IT workplace guideline. 1.2 Desktop **<u>Peripherals</u>** are checked and connected with computer properly. 1.3 Computer is switched on. 1.4 Computer **<u>desktop/GUI settings</u>** are arranged and customized as per requirement. 1.5 Files and folders are **<u>manipulated</u>** as per requirement. 1.6 Properties of files and folders are viewed and searched. 1.7 Disks are defragmented, formatted as per requirement.
2. Install application software	2.1 Installation requirements of software are identified and listed. 2.2 Software sources and CD key/ password are assured. 2.3 **<u>Appropriate Software</u>** are collected and selected as per requirement. 2.4 Software installation is started. 2.5 Customization is done as per requirement. 2.6 Steps of installation are followed as per installation Instructions. 2.7 Installations are completed properly. 2.8 Correctness of Installation is checked.
3. Use word processor to prepare/create documents	3.1 Appropriate **<u>word processor</u>** is selected and started. 3.2 Documents are created as per requirement in personal use and office environment. 3.3 Contents are entered. *3.4* Documents are formatted. 3.5 Paragraph and page settings are completed. 3.6 Document is saved.
4. Use spreadsheet to	4.1 **<u>Spreadsheet applications</u>** are selected and started.

create/prepare worksheets	4.2 Worksheets are created as per requirement in personal use and office environment. 4.3 Data are entered. 4.4 Functions are used for calculating and editing logical operation. 4.5 Sheets are formatted as per requirement. 4.6 Charts are created. 4.7 Charts/sheets are saved.
5. Use presentation software to create/ prepare presentation	5.1 Appropriate **presentation applications** are selected and started. 5.2 Presentations are created as per requirement in personal use and office environment 5.3 Image, illustrations, text, table, symbols and media are entered as per requirements. 5.4 Presentations are formatted and animated. 5.5 Presentations are viewed and saved.

Range of Variables

Variable	Range May include but not limited to:
1. Peripherals	1.1 Monitor 1.2 Keyboard 1.3 Mouse 1.4 Modem 1.5 Scanner 1.6 Printer
2. Desktop settings	2.1 Icons 2.2 Taskbar 2.3 View 2.4 Resolutions
3. Manipulate	3.1 Create 3.2 Open 3.3 Copy 3.4 Rename 3.5 Delete 3.6 Sort
4. Typing tutors	4.1 English typing tutor 4.2 Bangla typing tutor
5. Appropriate software	5.1 MS office or Open office but limited to 5.2 Word processor software. 5.3 Spread sheet software 5.4 Presentation software
6. Word processor	6.1 MS Word processor 6.2 Open office Org 6.3 Google docs 6.4 Word perfect 6.5 LibreOffice

7. Spread sheet applications	7.1 MS Excel 7.2 Google Sheets 7.3 Apple Numbers by Apple
8. Presentation application	8.1 MS PowerPoint 8.2 Google Slides 8.3 Prezi

Curricular Content Guide

1. Underpinning Knowledge	1.1 Desktop items 1.2 Type of Bangla keyboard layout 1.3 Different types of software and application packages 1.4 Use of word processor, spread sheet and presentation software 1.5 Types of printers 1.6 Types of charts, impotence of chart 1.7 Different types of math and logical functions
2. Underpinning Skills	2.1 Starting computer 2.2 Installing operating system 2.3 Managing desktop item 2.4 Manipulating files and folders as per requirement 2.5 Installing application software 2.6 Running application software 2.7 Creating and saving document with word processing application 2.8 Using functions in spread sheet. 2.9 Applying animations into presentation slide 2.10 Printing document
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Scanner, printer 4.4 Software 4.5 Work books 4.6 Relevant tools, equipment and facilities needed to perform the activities 4.7 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed operating system 1.2 manipulated files and folders as per requirement 1.3 Installed appropriate application software 1.4 used word processor to prepare/create documents 1.5 used spreadsheet to create/prepare worksheets 1.6 used presentation software to create/prepare presentation 1.7 applied animations into presentation slides 1.8 printed document
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **COMPLY WITH ETHICAL STANDARDS IN IT WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-ICT-MPS-02-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to comply with ethical standards in IT workplace. It specifically includes the tasks of upholding the requirements of clients, delivering quality products and services, maintaining professionalism at workplace and maintaining workplace code of conduct.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Uphold the requirements of clients	1.1 Clients' requirements are identified. 1.2 Confidentiality of information is maintained in accordance with workplace policies/organizational policies/national legislation. 1.3 Potential conflicts of interest are identified and involved parties of potential conflicts are notified. 1.4 Proprietary rights of client/customer are asserted.
2. Deliver quality products and services	2.1 Products and services are provided according to the clients' requirements. 2.2 Work is completed as per standards. 2.3 Quality processes are implemented when developing products and services.
3. Maintain professionalism at workplace	3.1 Work processes are delivered as per standards. 3.2 Skills, knowledge and qualifications are presented in a professional manner. 3.3 Services and products developed by self and others are delivered as per workplace standard. 3.4 Unbiased and objective information are provided to the clients. 3.5 Realistic estimates for time, cost and delivery of outputs are presented during negotiation.
4. Maintain workplace code of conduct	4.1 Workplace code of conduct is interpreted. 4.2 Workplace code of conduct is followed.

Range of Variables

Variable	Range May include but not limited to:

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Corporate code of confidentiality of information 1.2 Organizational policies, national legislation and workplace policies in relation to IT sector 1.3 Law and regulations pertaining to proprietary rights 1.4 Quality processes for products and services 1.5 Procedure of provided to client information 1.6 Method of estimating for time, cost and delivery products and services 1.7 Workplace code of conduct in IT sector
2. Underpinning Skills	2.1 Upholding confidentiality of information in accordance with organizational policies, national legislation and workplace policies 2.2 Asserting proprietary rights of clients/customers 2.3 Completing work in accordance with industry and international standards 2.4 Implementing quality processes when developing products and services 2.5 Delivering correctly services and products developed by self and others 2.6 Providing unbiased and objective information to the clients 2.7 Presenting realistic estimates for time, cost and delivery of outputs during negotiation 2.8 Maintaining workplace code of conduct
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 National legislation in relation to IT sector 4.4 Workplace policies, code of conduct & proprietary rights of clients in IT sector 4.5 Relevant tools, equipment and facilities needed to perform the activities 4.6 Required learning materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate:

	1.1 upheld the requirements of clients/customers 1.2 asserted proprietary rights of clients/customers 1.3 delivered quality products and services 1.4 Implemented quality processes when developing products and services 1.5 maintained professionalism at the workplace 1.6 completed work according to industry and international standards 1.7 maintained workplace code of conduct
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **RECOGNIZE THE BASIC ELECTRICAL AND ELECTRONIC CONCEPT**	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-MPS-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to recognize the basic electrical and electronics concept. It specifically includes the tasks of understanding basic electricity concept, identifying basic electronics components, understanding transformer functions, and identifying low powered cables.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand basic electricity concept	1.1 **Fundamental concepts of electricity** are understood and explained. 1.2 Relationship between current, voltage, and resistance is recognized. 1.3 Concept of frequency in alternating current (AC) systems is understood and explained. 1.4 **Basic electrical measurements** are recognized.
2. Identify basic electronic components	2.1 **Basic electronic components** are identified. 2.2 Functions and applications of electronic components are understood and explained. 2.3 Polarity and characteristics of diodes are recognized and used for proper circuit integration. 2.4 Functions and operation of NPN and PNP transistors is understood. 2.5 Capacitors are identified by their types and specifications. 2.6 Resistors are identified based on their resistance values and color codes.
3. Understand transformer functions	3.1 **Functions of transformers** are understood and explained. 3.2 Difference between step-up and step-down transformers is identified, and their applications are explained. 3.3 Primary and secondary windings of transformers are identified. 3.4 Use of transformers in electrical circuits to either increase or decrease voltage levels is understood.
4. Identify low powered cables	4.1 Low-powered cables used in mobile phone servicing are identified and categorized. 4.2 Function and specifications of low-powered cables are understood and explained.

	4.3 Differences between various low-powered cables are recognized and understood.

Range of Variables

Variable	**Range** May include but not limited to:
1. Fundamental concepts of electricity	1.1 Electric Charge 1.2 Current (I) 1.3 Voltage (V) 1.4 Resistance (R) 1.5 Ohm’s Law: V=I×RV = I \times RV=I×R 1.6 Power (P): P=V×IP = V \times IP=V×I 1.7 Circuits 1.8 Conductor, non-conductor/insulator 1.9 Semi-conductor
2. Basic electrical measurements	2.1 Voltage 2.2 Current 2.3 Resistance
3. Basic electronic components	3.1 Diodes 3.1.1 Zener diode 3.1.2 Schottky Diode 3.1.3 LED 3.2 Transistors (NPN and PNP) 3.3 Silicon Controlled Rectifier (SCR) 3.4 Capacitor (DC and AC) 3.5 Resistor 3.5.1 Variable 3.5.2 Fixed
4. Functions of transformers	4.1 Step-up voltage 4.2 Step-down voltage

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand and explain the fundamental concepts of electricity. 1.2 Recognize the relationship between current, voltage, and resistance. 1.3 Understand and explain the concept of frequency in alternating current (AC) systems. 1.4 Recognize basic electrical measurements. 1.5 Identify basic electronic components. 1.6 Understand and explain the functions and applications of electronic components. 1.7 Understand the functions and operation of NPN and PNP transistors. 1.8 Identify the difference between step-up and step-down transformers and explain their applications. 1.9 Understand the use of transformers in electrical circuits to

	increase or decrease voltage levels. 1.10 Understand and explain the function and specifications of low-powered cables.
2. Underpinning Skills	2.1 Performing basic electrical measurements and understanding key concepts of electricity. 2.2 Demonstrating the relationship between current, voltage, and resistance. 2.3 Applying the concept of frequency in AC systems for better understanding. 2.4 Fixing circuit issues by recognizing the polarity and characteristics of diodes. 2.5 Maintaining the operation of NPN and PNP transistors. 2.6 Recognizing resistors based on their resistance values and color codes. 2.7 Servicing transformers by understanding their functions and applications in electrical circuits.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Basic electronics components 4.3 Basic electrical components 4.4 Low-powered cables

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 understood basic electricity concept 1.2 identified basic electronics components 1.3 understood transformer functions 1.4 identified low powered cables
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY FUNDAMENTAL SKILLS FOR MOBILE PHONE WORKS**	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-MPS-02-O

Unit Descriptor:

This unit covers the knowledge, skills and attitudes required to apply fundamental skills for mobile phone works. It specifically includes the tasks of identifying basic concept of mobile phone, identifying typed of tools and equipment, identifying mobile phone servicing materials, identifying circuit and schematic diagram, measuring current and voltage and resistance using multimeter.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify basic concept of mobile phone	1.1 **Types of mobile phones** are identified based on their capabilities and uses. 1.2 **Mobile phone brands** are identified and recognized. 1.3 **Operating systems of mobile phones** are explained. 1.4 Version of mobile phones are recognized and based on their specifications.
2. Identify types of tools and equipment	2.1 **Basic hand tools** are identified and described. 2.2 **Diagnostic and testing equipment** are recognized and selected as per troubleshooting requirements. 2.3 **Soldering and desoldering tools** are identified and explained for use in board-level repairs. 2.4 **Anti-static tools** are identified and explained to prevent component damage.
3. Identify mobile phone servicing materials	3.1 **Common servicing materials** are identified based on their use in repair tasks. 3.2 **Cleaning materials** are identified for safe handling of mobile components. 3.3 Material specifications and compatibility are identified with the job requirement and manufacturer standards.
4. Identify circuit and schematic diagram	4.1 **Basic circuit diagrams** are identified and interpreted for mobile phone electronics. 4.2 Mobile phone schematic diagrams are identified and reviewed as per model specifications. 4.3 **Symbols used in circuit diagrams** are recognized and matched with components.
5. Measure current, voltage and resistance	5.1 Safety precautions are followed during all measurements to avoid damage or injury.

using multimeter	5.2 Multimeter is identified and set according to the required measurement type. 5.3 Voltage is measured across components or terminals using proper multimeter settings. 5.4 Current is measured by placing the multimeter in series with the circuit as per standard procedures. 5.5 Resistance is measured across components with the circuit powered off. 5.6 Measured values are recorded and compared with standard or expected readings.

Range of Variables

Variable	**Range** May include but not limited to:
1. Types of mobile phones	1.1 Basic phone 1.2 Features phone 1.3 Smart phone
2. Mobile phone brands	2.1 Samsung 2.2 MI (Xiaomi, Redmi, Realme) 2.3 Apple (I phone) 2.4 Vivo 2.5 Walton 2.6 Symphony 2.7 itel 2.8 Oppo 2.9 Infinix 2.10 Nokia
3. Operating systems of mobile phones	3.1 Android 3.2 iOS 3.3 Tizen 3.4 Microsoft windows
4. Basic hand tools	4.1 Screw driver 4.2 Tweezer 4.3 Casing opener 4.4 PCB holder 4.5 Nose pliers 4.6 Combination pliers 4.7 Cutting pliers 4.8 Blade cutter 4.9 Point cutter
5. Diagnostic and testing equipment	5.1 Avo meter/multimeter 5.1.1 Analog 5.1.2 Digital 5.2 DC Power Supply 5.3 GSM/3G/4G/5G Signal Tester 5.4 LCD Tester

6. Soldering and desoldering tools	6.1 Soldering stand 6.2 Soldering iron 6.3 PCB Holder 6.4 Tweezers
7. Anti-static tools	7.1 ESD Mat 7.2 ESD Wrist Strap 7.3 ESD Gloves 7.4 ESD Tweezer 7.5 ESD Shoes or Shoe Straps Ionizing
8. Common servicing materials	8.1 Soldering lead 8.2 Soldering paste 8.3 BGA/IC paste 8.4 Jumper wire 8.5 Heat tape 8.6 Both tape 8.7 Adhesive 8.8 OCA paper
9. Cleaning materials	9.1 De-soldering wick/Desoldering wire 9.2 Contact cleaner 9.3 Thinner 9.4 Iso propyl
10.Basic circuit diagrams	10.1 Power Supply Circuit 10.2 Charging Circuit 10.3 Display Circuit 10.4 Audio Circuit 10.5 Camera Circuit 10.6 Signal Circuit 10.7 Control Circuit 10.8 Touchscreen Circuit 10.9 Lighting Circuit
11.Symbols used in circuit diagrams	11.1 Battery 11.2 Resistor 11.3 Capacitor 11.4 Inductor 11.5 Diode 11.6 LED 11.7 Ground 11.8 Switch 11.9 Transistor (NPN and PNP) 11.10 Integrated Circuit (IC)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Explain the operating systems of mobile phones.

	1.2 Identify and describe basic hand tools. 1.3 Recognize and select diagnostic and testing equipment based on troubleshooting needs. 1.4 Identify and explain soldering and desoldering tools for board-level repairs. 1.5 Identify common servicing materials based on their use in repair tasks. 1.6 Identify and interpret basic circuit diagrams for mobile phone electronics. 1.7 Recognize symbols used in circuit diagrams and match them with components.
2. Underpinning Skills	2.1 Performing identification of mobile phone types based on capabilities and uses. 2.2 Demonstrating recognition of mobile phone brands and operating systems. 2.3 Applying knowledge of mobile phone versions based on specifications. 2.4 Fixing and describing basic hand tools for mobile phone repairs. 2.5 Maintaining diagnostic and testing equipment for troubleshooting tasks. 2.6 Servicing soldering and desoldering tools for board-level repairs. 2.7 Identifying and applying anti-static tools to prevent component damage. 2.8 Performing identification of servicing materials and cleaning materials for mobile components. 2.9 Applying safety precautions when measuring current, voltage, and resistance using a multimeter.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Mobile phone components 4.3 Basic hand tools 4.4 Diagnostic and testing equipment 4.5 Soldering and desoldering tools 4.6 Anti-static tools 4.7 Common servicing materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified basic concept of mobile phone 1.2 identified types of tools and equipment used in mobile phone servicing 1.3 identified the materials required in mobile phone servicing 1.4 identified circuit, schematic and block diagram 1.5 measured current, voltage and resistance
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IDENTIFY COMPONENTS OF MOBILE PHONE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-ICT-MPS-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to identify components of mobile phone. It specifically includes the tasks of identifying the sections of a mobile phone and identifying individual components and parts.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the sections of a mobile phone	1.1 Network sections are identified and described. 1.2 Power sections are identified and explained. 1.3 Control sections are identified are described. 1.4 Audio input and output sections are identified and explained. 1.5 Charging sections are identified and described. 1.6 Display sections are identified and explained.
2. Identify individual components and parts	2.1 **<u>Components and parts of the mobile phone</u>** are identified based on their function. 2.2 Components and parts of the mobile phone are listed systematically according to their category. 2.3 Function of each component is interpreted.

Range of Variables

Variable	Range May include but not limited to:
1. Components and parts of the mobile phone	1.1 Fuses 1.2 Inductor 1.3 Non-electrolytic Capacitor 1.4 Electrolytic capacitor 1.5 Resistors 1.6 Coupler 1.7 Sensor 1.8 Diode 1.9 LED 1.10 ICs (Processor, Power, Audio, lighting, Network, Wi-Fi-Bluetooth IC, Charging, Storage) 1.11 Receiver 1.12 Speaker 1.13 Charging Port 1.14 Headphone Port 1.15 Microphone

	1.16 Display Module 1.17 Camera Module 1.18 Keypad 1.19 Dome-sheet 1.20 Vibrator 1.21 Battery 1.22 Battery Connectors 1.23 SIM Card 1.24 Memory Card 1.25 SIM & Memory Base 1.26 SIM & Memory Tray 1.27 Switches 1.28 RX and TX Filter 1.29 Transistor 1.30 Antenna 1.31 Housing 1.31.1 A Housing 1.31.2 B Housing 1.31.3 C Housing

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and describe network sections of the mobile phone. 1.2 Explain power sections of the mobile phone. 1.3 Identify and describe control sections of the mobile phone. 1.4 Explain audio input and output sections. 1.5 Describe charging sections of the mobile phone. 1.6 Explain display sections of the mobile phone. 1.7 Identify components and parts of the mobile phone based on their function. 1.8 Interpret the function of each mobile phone component.
2. Underpinning Skills	2.1 Performing identification and description of network sections in a mobile phone. 2.2 Demonstrating identification and explanation of power sections in the mobile phone. 2.3 Applying knowledge to identify and describe control sections. 2.4 Fixing and explaining audio input and output sections in the mobile phone. 2.5 Maintaining identification and description of charging sections. 2.6 Servicing the identification and explanation of display sections. 2.7 Identifying components and parts of the mobile phone based on their function. 2.8 Listing components and parts systematically according to

	their category. 2.9 Interpreting the function of each mobile phone component.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Basic mobile phone components

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified the sections of a mobile phone 1.2 identified individual components and parts
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **DISASSEMBLE AND REASSEMBLE OF MOBILE PHONE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-ICT-MPS-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to disassemble and reassemble of mobile phone. It specifically includes the tasks of disassembling mobile phone, re-assembling the mobile phone and cleaning tools and equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Disassemble mobile phone	1.1 Safety precautions are followed to prevent damage to sensitive parts. 1.2 **<u>Tools</u>** are collected for disassemble the mobile phone. 1.3 Power is turned off, and the mobile phone is disconnected. 1.4 Screws, clips, and other fasteners are removed carefully without damaging components. 1.5 Internal components are carefully detached using proper techniques. 1.6 Components are organized and stored to prevent damage or misplacement during disassembly.
2. Re-assemble the mobile phone	2.1 Safety precautions are followed during reassembly, including ESD protection. 2.2 Tools are collected for reassemble the mobile phone. 2.3 All components are checked for proper functionality and alignment before reassembly. 2.4 Internal components are carefully reconnected using proper techniques. 2.5 Screws, clips, and fasteners are reattached without over-tightening or damaging components. 2.6 Reassembled mobile phone is powered on and tested to ensure proper functionality.
3. Clean tools and equipment	3.1 Tools and equipment are checked for any signs of wear, damage, or contamination before cleaning. 3.2 Tools and equipment are cleaned thoroughly to remove dirt, grease, or other residues. 3.3 Cleaned tools and equipment are dried and stored in an organized manner for future use.

Range of Variables

Variable	Range May include but not limited to:
1. Tools	1.1 Screw driver 1.2 Casing opener 1.3 Tweezer 1.4 Nose pliers 1.5 Point cutter 1.6 LCD separator/Pre heater

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify necessary tools and safety equipment for disassembly. 1.2 Describe the proper techniques for removing screws, clips, and fasteners. 1.3 Electrostatic Discharge (ESD) protection protocols during reassembly. 1.4 Interpret all components are aligned and functioning correctly before reassembly. 1.5 Procedures of clean tools and equipment.
2. Underpinning Skills	2.1 Demonstrating collection of tools for disassembling the mobile phone. 2.2 Applying the procedure to turn off power and disconnect the mobile phone. 2.3 Servicing the organization and storage of components to prevent damage or misplacement. 2.4 Collecting tools for reassembling the mobile phone. 2.5 Testing the reassembled mobile phone for proper functionality. 2.6 Checking tools and equipment for signs of wear, damage, or contamination before cleaning. 2.7 Cleaning tools and equipment thoroughly to remove dirt, grease, or residues. 2.8 Drying and storing cleaned tools and equipment in an organized manner for future use.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools

	4.3 Mobile phone

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 disassembled mobile phone 1.2 re-assembled the mobile phone 1.3 cleaned tools and equipment
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PRACTICE SOLDERING AND DE SOLDERING**	**Nominal Duration:** 60 hrs.	**Unit Code:** SICIP-ICT-MPS-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to practice soldering and resoldering. It specifically includes the tasks of selecting soldering tools and materials, performing soldering on PCB, performing de-soldering components from PCB, performing re-soldering from PCB, and cleaning soldering tools and workspace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select soldering tools and materials	1.1 **<u>Soldering tools</u>** are selected based on the specific repair task requirements. 1.2 **<u>Soldering materials</u>** are selected according to the type of components being soldered. 1.3 Soldering tools are used according to safety guidelines to prevent damage.
2. Perform soldering on PCB	2.1 Safety precautions are followed during soldering to avoid component damage. 2.2 PCB components are positioned and secured in place to ensure correct alignment before soldering. 2.3 Soldering iron is heated to the appropriate temperature, 2.4 Soldering is carried out on the PCB pads and component leads. 2.5 Solder joints are inspected for quality. 2.6 Excess flux is cleaned from the PCB after soldering to prevent corrosion or damage.
3. Perform de-soldering components from PCB	3.1 Safety measures are followed, ensuring adequate ventilation. 3.2 Component leads are heated with the soldering iron to melt the solder and prepare for removal. 3.3 Soldering hot air gun is applied to remove the molten solder and safely detach the component. 3.4 Component is carefully removed without causing damage to the PCB. 3.5 Excess solder and residual flux are cleaned from the PCB after component removal.
4. Perform re-soldering from PCB	4.1 PCB is inspected for faulty solder joints or components that require re-soldering. 4.2 Faulty component or joint is heated. 4.3 Old solder is removed using solder wick. 4.4 New solder is applied to the PCB. 4.5 Re-soldered joint is inspected for strength, proper connection, and any signs of cold solder joints or shorts.

	4.6 PCB is cleaned of any flux residue after re-soldering.
5. Clean soldering tools and workspace	5.1 Soldering tools are cleaned using appropriate cleaning materials. 5.2 Soldering iron tips are wiped clean and stored properly to avoid oxidation. 5.3 Workspace is cleared of any excess materials.

Range of Variables

Variable	**Range** May include but not limited to:
1. Soldering tools	1.1 Soldering stand 1.2 Soldering iron 1.3 PCB Holder 1.4 Tweezers 1.5 Point cutter 1.6 Nose pliers 1.7 ESD mat
2. Soldering materials	2.1 Soldering paste 2.2 Soldering lead 2.3 IC paste 2.4 Glue 2.5 Jumper wire 2.6 Heat tape 2.7 Both tape

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify soldering tools based on the specific repair task requirements. 1.2 Describe soldering materials according to the type of components being soldered. 1.3 Describe position and secure PCB components to ensure correct alignment before soldering. 1.4 Heating procedures the soldering iron to the appropriate temperature for the task. 1.5 Inspecting procedures of solder joints for quality and proper formation. 1.6 Explain the adequate ventilation by following safety measures during soldering. 1.7 Soldering iron tips clean and store properly to avoid oxidation.
2. Underpinning Skills	2.1 Selecting soldering tools and materials based on specific repair task requirements. 2.2 Demonstrating the selection of soldering materials according to component types. 2.3 Applying soldering tools according to safety guidelines to prevent damage.

	2.4 Performing soldering on PCB, ensuring safety precautions are followed. 2.5 Fixing PCB components in place to ensure correct alignment before soldering. 2.6 Performing de-soldering by following safety measures and using the proper tools. 2.7 Re-soldering faulty joints by heating, removing old solder, and applying new solder. 2.8 Inspecting re-soldered joints for strength, connection, and defects.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Soldering tools 4.3 Soldering materials 4.4 PCB board

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 selected soldering tools and materials 1.2 performed soldering on PCB 1.3 performed de-soldering components from PCB 1.4 performed re-soldering from PCB 1.5 cleaned soldering tools and workspace
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **TEST COMPONENT OF MOBILE PHONE**	**Nominal Duration:** 60 hrs.	**Unit Code:** SICIP-ICT-MPS-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to test component of mobile phone. It specifically includes the tasks of understanding standard testing criteria for each component, selecting correct tools and testing method and testing components in-circuit or off-circuit.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand standard testing criteria for each component	1.1 **<u>Standard testing criteria</u>** for each electronic component are understood. 1.2 Each component's specifications are understood to ensure proper testing.
2. Select correct tools and testing method	2.1 **<u>Testing tools</u>** is identified. 2.2 **<u>Appropriate testing methods</u>** for each component are recognized. 2.3 Testing method is chosen to match the type of component. 2.4 Testing methods are followed to ensure that the results align with component specifications.
3. Test components in-circuit or off-circuit	3.1 Testing method is selected based on the type of component being tested. 3.2 In-circuit testing is performed. 3.3 Off-circuit testing is performed. 3.4 Test results are interpreted to determine if the component functions correctly. 3.5 Corrective actions are carried out as needed after testing.

Range of Variables

Variable	**Range** May include but not limited to:
1. Standard testing criteria	1.1 Verify component values (resistance, capacitance, inductance) within tolerance. 1.2 Ensure voltage, current, and power ratings are within limits. 1.3 Test for leakage currents and thermal stability. 1.4 Check switching speed and reliability under stress.

	1.5 Perform functional and lifespan testing for durability. 1.6 Ensure compliance with safety and environmental standards.
2. Testing tools	2.1 Avo meter/multimeter 2.1.1 Analog 2.1.2 Digital 2.2 DC Power Supply 2.3 GSM/3G/4G/5G Signal Tester 2.4 LCD Tester
3. Appropriate testing methods	3.1 Resistance test 3.2 Continuity test 3.3 Voltage test

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand standard testing criteria for each electronic component. 1.2 Comprehend each component's specifications to ensure proper testing. 1.3 Identify appropriate testing tools for component testing. 1.4 Recognize appropriate testing methods for each component. 1.5 Testing method that matches the type of component. 1.6 Identify in-circuit testing. 1.7 Identify off-circuit testing. 1.8 Interpret test results to determine if the component functions correctly.
2. Underpinning Skills	2.1 Understanding standard testing criteria for each electronic component. 2.2 Demonstrating understanding of each component's specifications for proper testing. 2.3 Applying knowledge to identify correct testing tools. 2.4 Fixing the testing method to match each component's requirements. 2.5 Maintaining testing methods to ensure alignment with component specifications. 2.6 Performing in-circuit testing based on the component type. 2.7 Conducting off-circuit testing as necessary. 2.8 Servicing test results by interpreting them to ensure proper component functionality. 2.9 Performing corrective actions as needed after testing results.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Testing tools

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 understood standard testing criteria for each component 1.2 selected correct tools and testing method 1.3 tested components in-circuit or off-circuit
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE MOBILE PHONE**	Nominal Duration: 80 hrs.	Unit Code: SICIP-ICT-MPS-07-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to service mobile phone. It specifically includes the tasks of preparing for servicing works, disassemble mobile phone, testing mobile phone components, replacing faulty components, re-assembling mobile phone, and cleaning tools and equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for servicing works	1.1 Mobile phone is inspected thoroughly to assess the type of servicing required. 1.2 **All necessary tools and equipment** are gathered and checked for proper functionality. 1.3 Service area is prepared by ensuring it is clean, and organized. 1.4 Repair service manual is identified to ensure all necessary steps are followed and completed.
2. Disassemble mobile phone	2.1 Mobile phone is carefully examined to identify the components that need to be disassembled. 2.2 Mobile phone is powered off. 2.3 Screws, clips, and adhesives securing the phone's outer casing are identified and removed. 2.4 Back cover, screen, or other components are gently separated. 2.5 All internal components are carefully disconnected. 2.6 Disassembled parts are organized and placed in a safe, clean area to avoid loss or damage.
3. Test mobile phone components	3.1 Battery is tested for voltage and charging capacity using appropriate tools, such as a multimeter. 3.2 Display and touch functionality are tested by powering on the device and checking for responsiveness. 3.3 **Buttons** are tested to ensure they function correctly. 3.4 Speakers, microphone, and audio ports are tested by playing sound and recording audio. 3.5 Cameras are tested by taking photos and recording videos to ensure proper focus and image quality. 3.6 **Connectivity features** are tested.
4. Replace faulty components	4.1 Faulty components are identified through testing or inspection. 4.2 Necessary replacement parts are selected. 4.3 Faulty component is carefully removed without damaging surrounding parts.

	4.4 Replacement component is checked for compatibility with the mobile phone model and prepared for installation. 4.5 New component is installed and securely connected to the relevant parts.
5. Re-assemble mobile phone	5.1 Mobile phone components are carefully reassembled following the manufacturer's guidelines. 5.2 Internal components are reconnected securely, ensuring proper alignment and connection. 5.3 Screen, back cover, or any external parts are carefully positioned and reattached. 5.4 All screws, clips, and fasteners are tightened or secured to ensure the phone is fully assembled. 5.5 Mobile phone is powered on to verify that all components are functioning correctly after reassembly. 5.6 The phone's buttons, screen, ports, and other external features are tested.
6. Clean tools and equipment	6.1 Tools and equipment are inspected to check for any dirt, dust, or residue. 6.2 Appropriate cleaning materials are selected based on the type of tool or equipment. 6.3 Tools and equipment are cleaned thoroughly by wiping or scrubbing to remove any dirt, grease, or contaminants. 6.4 Testing equipment is cleaned using appropriate techniques. 6.5 After cleaning, tools and equipment are stored properly.

Range of Variables

Variable	**Range** May include but not limited to:
1. All necessary tools and equipment	1.1 Screw driver 1.2 Tweezer 1.3 Casing opener 1.4 PCB holder 1.5 Nose pliers 1.6 Combination pliers 1.7 Cutting pliers 1.8 Blade cutter 1.9 Point cutter 1.10 Soldering stand 1.11 Soldering iron 1.12 PCB Holder 1.13 Tweezers
2. Buttons	2.1 Power 2.2 Volume 2.3 Home buttons
3. Connectivity features	3.1 Wi-Fi 3.2 Bluetooth

	3.3 GPS 3.4 Network

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify all necessary tools and equipment and check them for proper functionality. 1.2 Identify the repair service manual to ensure all necessary steps are followed. 1.3 Identify components that need disassembly. 1.4 Identify screws, clips, and adhesives securing the phone's outer casing. 1.5 Identify the back cover, screen, or other components. 1.6 Explain battery for voltage and charging capacity using appropriate tools. 1.7 Describe display and touch functionality by powering on the device and checking responsiveness. 1.8 Identify faulty components through testing or inspection.
2. Underpinning Skills	2.1 Performing thorough inspection of the mobile phone to assess servicing needs. 2.2 Demonstrating the gathering and checking of tools and equipment for proper functionality. 2.3 Applying proper organization and cleanliness to the service area. 2.4 Fixing the repair service manual to ensure all necessary steps are followed. 2.5 Performing careful disassembly of the mobile phone. 2.6 Testing mobile phone components, including battery, display, buttons, speakers, microphone, and cameras. 2.7 Servicing faulty components by identifying, removing, and replacing them with compatible parts. 2.8 Reassembling the mobile phone by following manufacturer guidelines and securely connecting components. 2.9 Inspecting tools and equipment for dirt, dust, or residue and cleaning them thoroughly.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Tools and equipment 4.3 Mobile phone 4.4 Mobile phone parts

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 prepared for servicing works 1.2 disassembled mobile phone 1.3 tested mobile phone components 1.4 replaced faulty components 1.5 re-assembled mobile phone 1.6 cleaned tools and equipment
2. Methods of Assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY SOFTWARE INSTALLATION AND FLASHING MOBILE PHONE**	**Nominal Duration:** 40 Hrs**.**	**Unit Code:** SICIP-ICT-MPS-08-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply software installation and flashing mobile phone. It specifically includes the tasks of identifying software requirements of mobile phone, preparing flashing tools and install drivers, connecting mobile phone to PC or flashing tool, flashing mobile phone software and cleaning tools and equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify software requirements of mobile phone	1.1 Software requirements for installing and flashing the mobile phone are identified. 1.2 Appropriate flashing tools and software are identified based on the mobile phone brand and model. 1.3 Mobile phone's model and specifications are checked. 1.4 Required version of the mobile phone's operating system or firmware is identified and verified. 1.5 System requirements for flashing are checked and confirmed.
2. Prepare flashing tools and install drivers	2.1 **<u>Necessary flashing tools</u>** are identified and gathered based on the mobile phone model and software requirements. 2.2 Flashing tool is installed on the computer. 2.3 Correct version of flashing software is selected. 2.4 Drivers required for the flashing process are identified and downloaded from reliable sources. 2.5 Drivers are installed on the computer. 2.6 Installation of flashing tools and drivers is confirmed and checked.
3 Connect mobile phone to PC or flashing tool	3.1 Mobile phone is powered off before connecting it to the PC or flashing tool to avoid any electrical damage. 3.2 Appropriate USB cable is selected. 3.3 Mobile phone is connected to the PC using the USB cable. 3.4 Flashing tool or software is launched on the PC, and the mobile phone is detected in the tool's interface. 3.5 Any necessary drivers are installed on the PC. 3.6 PC or flashing tool is configured to detect the connected mobile phone, and the connection is tested for stability.
4 Flash mobile phone software	4.1 Mobile phone is prepared by ensuring it is fully charged, in the correct mode. 4.2 Fashing tool is configured to the appropriate settings,

	including the correct partition and flashing method. 4.3 Mobile phone is detected by the flashing software, and any necessary drivers are verified. 4.4 Mobile phone flashing is carried out as requirement. 4.5 Once the flashing process is completed, the mobile phone is automatically rebooted. 4.6 Mobile phone is tested to confirm that the new software is functioning correctly.
5 Clean tools and equipment	5.1 Tools and equipment are inspected to ensure they are free of dirt, debris, or residue. 5.2 Appropriate cleaning materials are selected based on the type of tool or equipment. 5.3 Tools and equipment are carefully cleaned to remove any dirt, grease, or contaminants. 5.4 Work area is cleaned to prevent tools or equipment from becoming contaminated again. 5.5 Cleaned tools and equipment are stored.

Range of Variables

Variable	**Range** (May includes but not limited to):
1. Necessary flashing tools	1.1 Odin flash tool 1.2 SP flash tool 1.3 SPD flash tool 1.4 Qualcomm flash tool 1.5 MTK flash tool

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify software requirements for installing and flashing the mobile phone. 1.2 Identify appropriate flashing tools and software based on the mobile phone brand and model. 1.3 Identify mobile phone's model and specifications. 1.4 system requirements for flashing. 1.5 Identify necessary flashing tools based on mobile phone model and software requirements. 1.6 Identify necessary drivers for the flashing process from reliable sources. 1.7 Installation procedure of flashing tools and drivers.
2. Underpinning Skills	2.1 Performing identification of software requirements for installing and flashing the mobile phone. 2.2 Fixing the required version of the operating system or firmware and verifying system requirements. 2.3 Maintaining organization of flashing tools and installation of required drivers on the computer. 2.4 Servicing the connection of the mobile phone to the PC or flashing tool, ensuring proper configuration.

	2.5 Performing flashing of mobile phone software by configuring the flashing tool to the correct settings. 2.6 Monitoring the flashing process to ensure no errors occur and tracking progress until completion. 2.7 Cleaning tools and equipment to remove dirt and contaminants, ensuring they are stored properly.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Mobile phone 4.4 Flashing tools 4.5 Different type software 4.6 necessary drivers

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified software requirements of mobile phone 1.2 prepared flashing tools and install drivers 1.3 connected mobile phone to PC or flashing tool 1.4 flashed mobile phone software 1.5 cleaned tools and equipment
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Mobile Phone Servicing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)

  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/ Laptop with software	05
2.	Multimedia Projector	01
3.	Working Station	25
4.	Hot Air Rework Stations	25
5.	Soldering Stations	25
6.	Soldering Irons	25
7.	Multimeters	25
8.	Power Supply Units	25
9.	Battery Testers	05
10.	Screen Testing Jigs (for LCD and touchscreens)	01
11.	USB Testers (for current and voltage checks)	05
12.	Screwdrivers Set (Precision set, including Phillips and flathead)	25
13.	Tweezers (ESD-safe, various sizes)	25
14.	Opening Tools	25
15.	SIM Card Ejector Pins	25
16.	Suction Cups (for screen removal)	25
17.	Magnifying Glasses with Lamp	25
18.	Defective Mobile Phone	15
19.	New Mobile Phone	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Mobile Phone Servicing, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.